Intelligent Sports Analytics: A Progressive Web App for Extracting and Evaluating Team Results

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Hockey | Table Tennis | Soccer

ELO Rating

Rating from: Last 5 Years

teamname	Elo Overall last 5 Years
FC Bayern München	1741.68
Bayer Leverkusen	1708.94
Borussia Dortmund	1651.21
RB Leipzig	1633.25
VfB Stuttgart	1571.85
Eintracht Frankfurt	1556.57
SC Freiburg	1542.67
1899 Hoffenheim	1491.18
1. FC Heidenheim 1846	1489.26
VfL Wolfsburg	1482.88
Bor. Mönchengladbach	1476.46
1. FC Union Berlin	1471.22
VfL Bochum	1455.6
SV Werder Bremen	1450.59
FC Augsburg	1447.54
1. FC Köln	1436.43
1. FSV Mainz 05	1419.98
Arminia Bielefeld	1419.77
SV Darmstadt 98	1406.49
FC Schalke 04	1395.54
SpVgg Greuther Fürth	1385.49
Hertha BSC	1365.39

Win Percentage Rating

Rating from: Last 5 Years

teamname	Elo Percent last 5 Years
FC Bayern München	76.83
Borussia Dortmund	67.73
Bayer Leverkusen	63.22
RB Leipzig	62.74
Eintracht Frankfurt	60.61
1. FC Union Berlin	58.68
SC Freiburg	58.66
VfL Wolfsburg	53.16
Bor. Mönchengladbach	45.85
1. FSV Mainz 05	44.92
1. FC Heidenheim 1846	44.27
VfB Stuttgart	44.26
1899 Hoffenheim	44.02
1. FC Köln	41.69
VfL Bochum	40.55
FC Augsburg	39.11
Arminia Bielefeld	35.79
SV Werder Bremen	34.93
Hertha BSC	34.83
FC Schalke 04	23.51
SV Darmstadt 98	22.02
SpVgg Greuther Fürth	20.12

Next 12 Games with ELO prediction

1. FC Heidenheim 1846	VS	Borussia Dortmund
9.92%	13.22%	76.86%
10.08	7.56	1.30
	02/02/2024	

SC Freiburg	VS	VfB Stuttgart
25.65%	34.21%	40.14%
3.90	2.92	2.49
	03/02/2024	

1. FSV Mainz 05	VS	SV Werder Bremen
25.44%	33.93%	40.63%
3.93	2.95	2.46
	03/02/2024	

VfL Bochum	VS	FC Augsburg
32.82%	38.39%	28.79%
3.05	2.60	3.47
	03/02/2024	

Next 12 Games with Win Percentage prediction

1. FC Heidenheim 1846	VS	Borussia Dortmund
15.92%	21.23%	62.84%
6.28	4.71	1.59
	02/02/2024	

SC Freiburg	VS	VfB Stuttgart
50.16%	28.48%	21.36%
1.99	3.51	4.68
	03/02/2024	

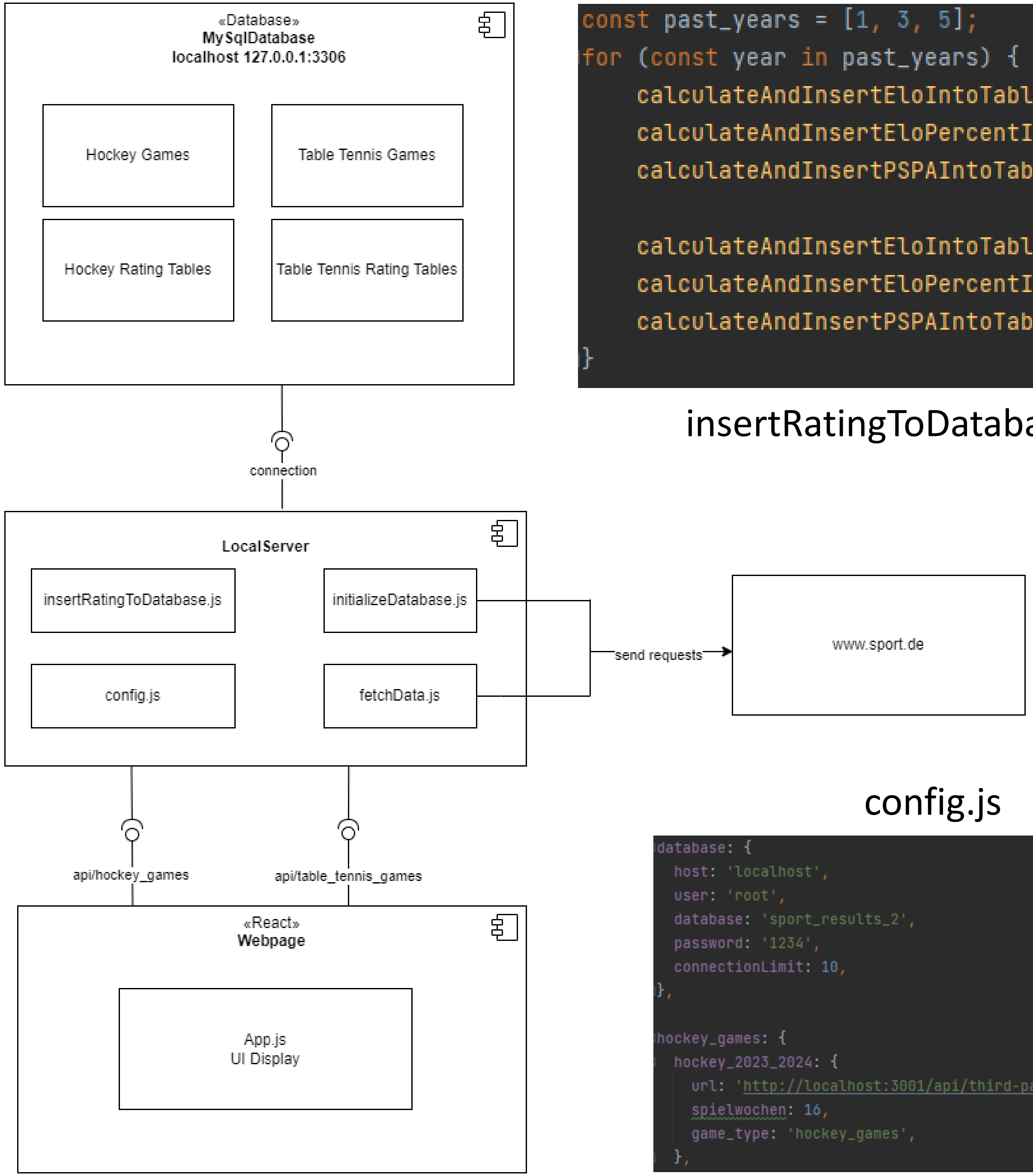
1. FSV Mainz 05	VS	SV Werder Bremen
43.99%	32.01%	24.01%
2.27	3.12	4.17
	03/02/2024	

VfL Bochum	VS	FC Augsburg
32.02%	38.85%	29.14%
3.12	2.57	3.43
	03/02/2024	

1. Project Definition

- provide a Web-User Interface for hockey, table tennis and soccer
- give an overview about three different ratings and predictions of the teams
- option to evaluate past games on the past predicted ratings.
- two main goals:
- predict future games
- evaluate past games

2. System Architechture




```
const past_years = [1, 3, 5];
for (const year in past_years) {
    calculateAndInsertEloIntoTable( sport: "hockey
    calculateAndInsertEloPercentIntoTable( sport:
    calculateAndInsertPSPAIntoTable( sport: "hocke

    calculateAndInsertEloIntoTable( sport: "hockey
    calculateAndInsertEloPercentIntoTable( sport:
    calculateAndInsertPSPAIntoTable( sport: "hocke
}
```

insertRatingToDatabase.js

config.js

```
database: {
  host: 'localhost',
  user: 'root',
  database: 'sport_results_2',
  password: '1234',
  connectionLimit: 10,
},

hockey_games: {
  hockey_2023_2024: {
    url: 'http://localhost:3001/api/third-party-data/?url=https:/
    spielwochen: 16,
    game_type: 'hockey_games',
  },
```

3.1. Elo Rating

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}$$

E_A ... Expected rating Team A
R_A ... Rating Team A
R_B ... Rating Team B

$$R_{Anew} = R_A + k * (outcome - E_A)$$

R_{Anew} ... New Rating Team A
k ... adaptive konstant k
$outcome$ ... [1, 0.5, 0] = [win , draw, lose]

3.2. Win Percentage Rating

- take the win rate of a team over a specific time period
- adjust it whether the team faced „good“ or „bad“ opponents.
- Example for a Team A facing 2 different teams.

Team A 2 : 0 Team B

Team A 0 : 1 Team C

Team A = 50%	League Average = 50%	Team A faced
Team B = 40%	Team B = -10%	Teams -15% below
Team C = 30%	Team C = -20%	Team A new = 35%

3.3. Points Scored/Points Allowed Rating

- Very useful measure since it can calculate win rates
- total points scored / points allowed over a time period
- devide through the number of games
- adjust like in win percentage rating

Team A 3 : 0 Team B

Team A 0 : 1 Team C

Team A PS/G = 1.5

Team A PA/G = 0.5

4.1 Elo Prediction

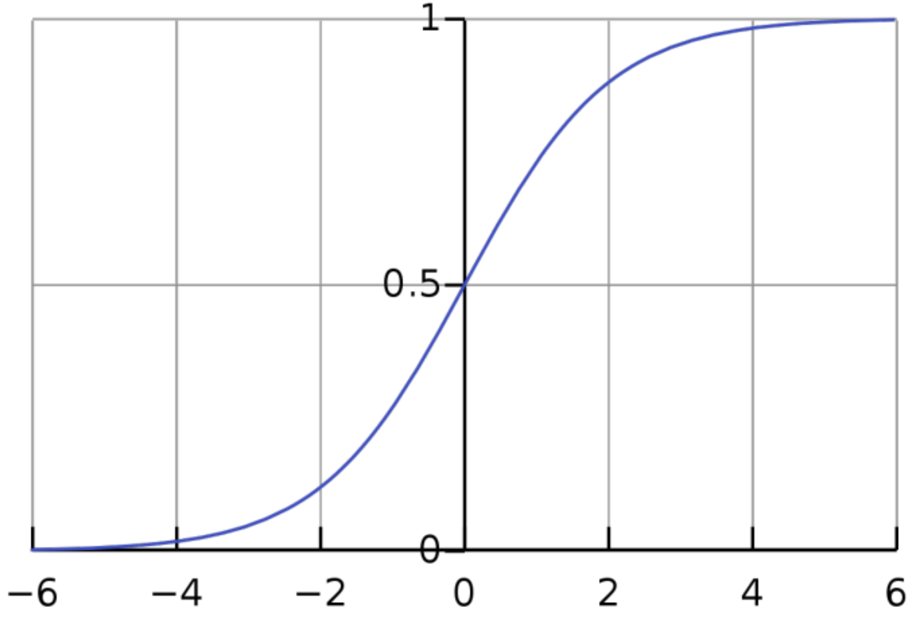
$$sigmoid(x) = \frac{1}{1+e^{-x}}$$

$$diff_{A-B} = R_A - R_B$$

$$diff_{B-A} = R_B - R_A$$

$$TeamA_{win} = sigmoid(\frac{diff_{A-B}}{spread})$$

$$TeamB_{win} = sigmoid(\frac{diff_{B-A}}{spread})$$



spread = 100

4.2 Win Percentage Prediction

- difference between two teams win percentage rating
- Evenly distrubute it around 50%
- If difference >50 threshold it
- Example:

Team A = 70%

Team B = 50%

Diff = 20%

Team A 70% : 30% Team B

4.3. Points Scored/Points Allowed Prediction

- Pythagorean Win Expectancy
- Exponent depends on goals
- Log5 Win Probability

$$Team_{win} = \frac{PS^{exp}}{PA^{exp} + PS^{exp}}$$

$$TeamA_{winning} = \frac{TeamA_{win} - (TeamA_{win} * TeamB_{win})}{TeamA_{win} + TeamB_{win} - 2 * TeamA_{win} * TeamB_{win}}$$

$TeamA_{winning}$...The probability that Team A beats Team B
$TeamA_{win}$...Pythagorean Win Expectancy of Team A
$TeamB_{win}$...Pythagorean Win Expectancy of Team B

5. Draw & Betting Quotes

- results are not binary
- possibility of a draw
- draw more likely the more evenly teams are matched

$$diff = |TeamA_{win} - TeamB_{win}|$$

$$draw = (1 - diff) * k$$

k ... constant

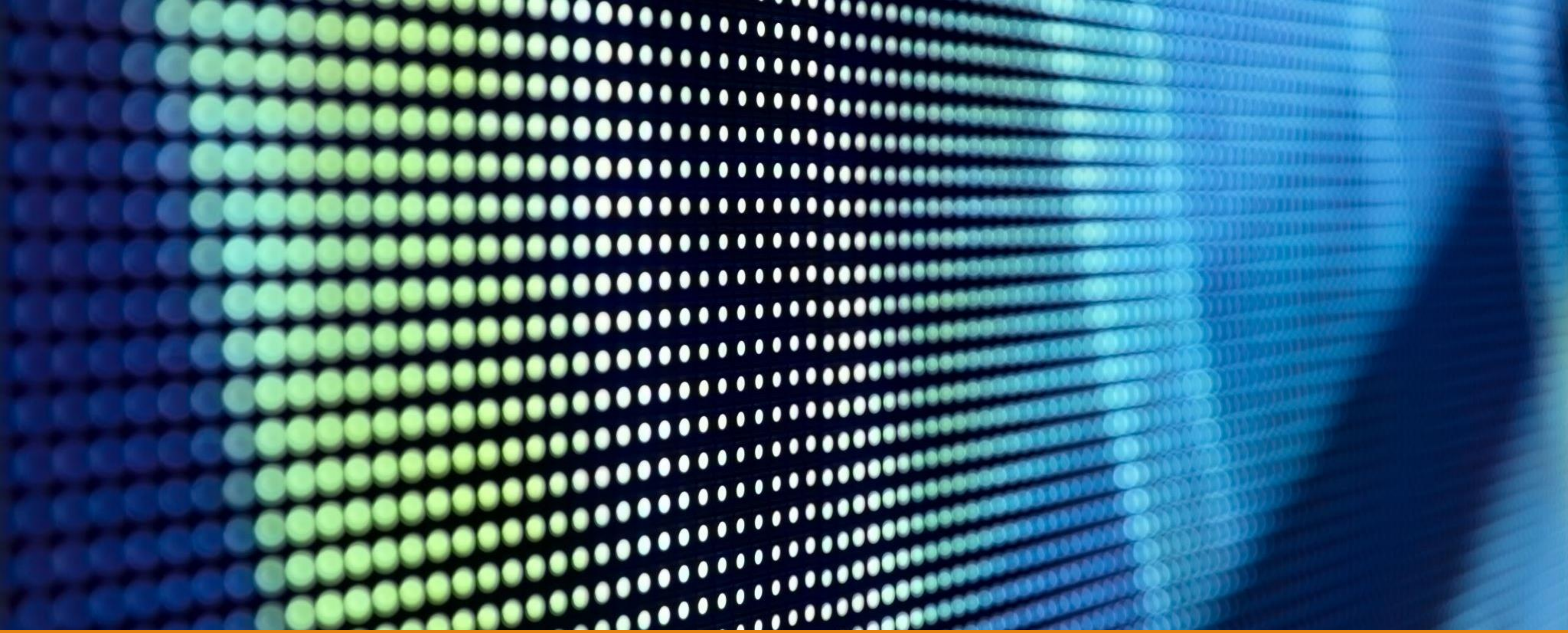
$$TeamA_{win} = TeamA_{win} - \frac{draw}{2}$$

$$TeamB_{win} = TeamB_{win} - \frac{draw}{2}$$

$$TeamA_{Quote} = \frac{1}{TeamA_{win}}$$

$$TeamB_{Quote} = \frac{1}{TeamB_{win}}$$

$$draw_{Quote} = \frac{1}{draw}$$



6. Live Demo

7. Q & A